NAVYA NANDAN

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PROFILE SUMMARY

Final-year B.Tech student in Electrical and Electronics Engineering with strong foundations in embedded systems, control engineering, and circuit design. Experienced with MATLAB, Simulink, Arduino, and microcontrollers. Passionate about developing efficient embedded and automation solutions with a focus on hardware-software integration and real-time systems. Quick learner with excellent problem-solving and debugging skills.

ACADEMIC QUALIFICATION

- Bachelor of Technology – **Electrical and Electronics Engineering**, Mar Athanasius College of Engineering, Kothamangalam
  CGPA 7.92 out of 10 | June, 2026
- Senior Secondary (**Class 12)** - Thrissur
  Percentage 93% | March, 2020
- Secondary Education (**Class 10**) - Thrissur
  Percentage 98% | March 2018

PROJECTS

Hybrid Control Prosthetic Finger | April 2026

- Developed a prosthetic finger using flex sensors and ESP32 for real-time gesture-based control.
- Integrated MG90S servo motors and Voice Recognition Module V3 for hybrid manual and voice operation.
- Tools & Technologies: ESP32, Arduino IDE, Embedded C, Flex Sensors, Servo Motors, LM393 Comparator.
- Demonstrates embedded systems, biomedical engineering, and human-machine interaction.

Regenerative Braking System for Electric Vehicles | March 2025

- Designed and simulated a regenerative braking system to convert kinetic energy into electrical energy during deceleration.
- Tools: MATLAB, Simulink, Arduino
- Achieved ~15% simulated energy recovery efficiency.

Arduino-Based Text-to-Speech Converter | October 2024

- Developed a real-time embedded system that converts digital text input to audible speech using Arduino and speech synthesis.
- Tools: Arduino IDE, C, Speaker Module
- Enhanced accessibility applications for visually impaired users.

Digital Fuel Gauge | April 2024

- Built a microcontroller-based system to display real-time fuel levels using sensors and digital readouts.
- Tools: Arduino, Ultrasonic Sensor, LCD Display

ACHIEVEMENTS

Received Best Project Award in the department for the project Hybrid Control Prosthetic Finger.

PUBLICATIONS

Published a paper titled *"Hybrid Control Prosthetic Finger: Underactuated Biomimetic Design With Voice And Flex Integration"* in IJCRT, Vol. 14, Issue 3, 2026.

CORE COMPETENCIES

- **Technical:** Python, C, MATLAB, Simulink, Arduino IDE, Proteus
- **Soft Skills:** Leadership, Team Work & Collaboration, Adaptability

INTERNSHIP

Student Trainee, **KSEB Electrical Section,** Kuruppampady| June 2024
Completed a field internship at the Kerala State Electricity Board, where I participated in AB switch installation and gained hands-on exposure to substation operations and power distribution systems. The experience strengthened my understanding of electrical networks, safety practices, and the practical application of technical problem-solving — skills valuable in both engineering and IT domains.

WORKSHOPS &TECHNICAL EVENTS

- Attended a Workshop on Embedded Systems in Electric Vehicles, gaining insights into control circuits, sensors, and energy management in EV technology.
- Participated in a Workshop on Electrical and Hybrid Vehicles, understanding powertrain systems and regenerative braking concepts.

CERTIFICATIONS & COURSES

- **NPTEL – The Joy of Computing Using Python** | 2024
  Completed a certified course covering Python programming fundamentals and problem-solving logic.
- **MathWorks – Machine Learning Onramp** | 2025.
- **Spoken Tutorial, IIT Bombay – Python** | 2024
- **Spoken Tutorial, IIT Bombay – C Programming** | 2024

AREA OF INTEREST

- Programming in Python and C, Embedded Systems and Internet of Things (IoT), Control Systems and Automation, Electric and Hybrid Vehicles, Machine Learning Applications in Engineering

LANGUAGES KNOWN

- English (Fluent) | Malayalam (Native) | Hindi (Basic) | Sanskrit (Basic)